



COMPUTER NETWORKS

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Department of Computer Science and Engineering

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Application Layer

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Unit – 2 Application Layer

2.3 The Domain Name System

2.4 P2P Applications

2.5 Socket Programming with TCP & UDP

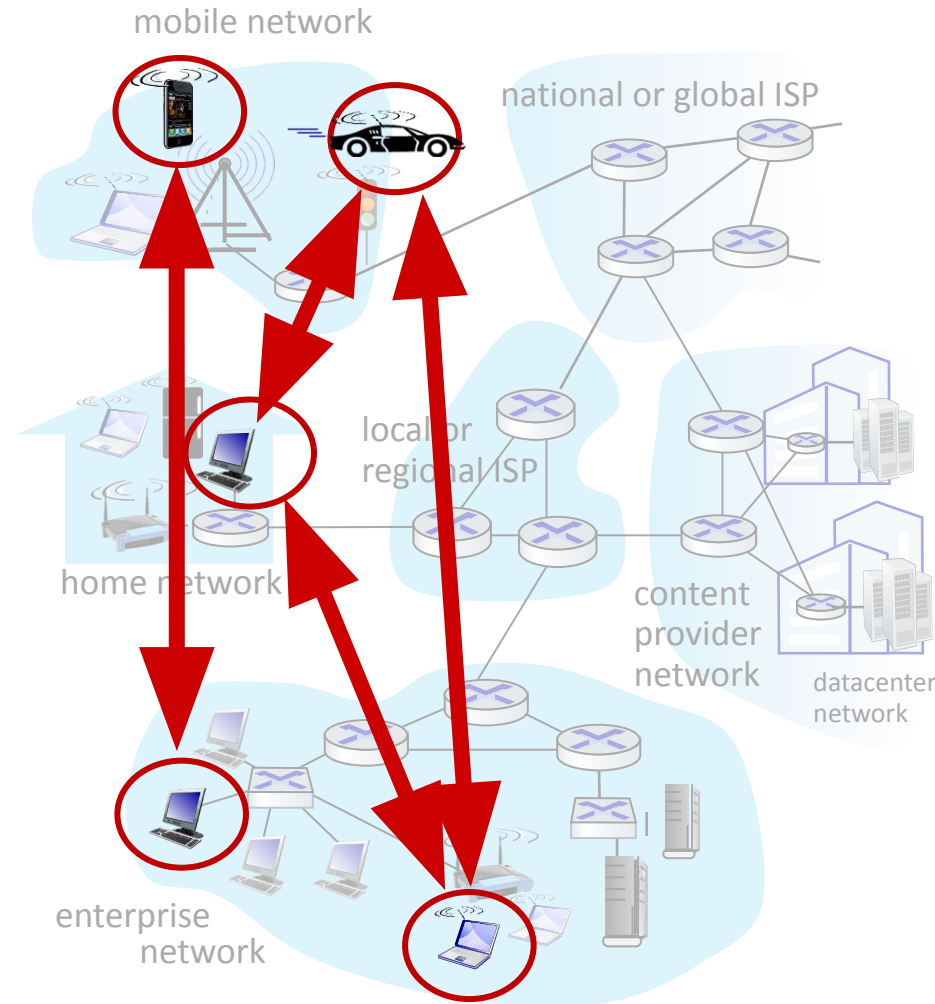
2.6 Other Application Layer Protocols



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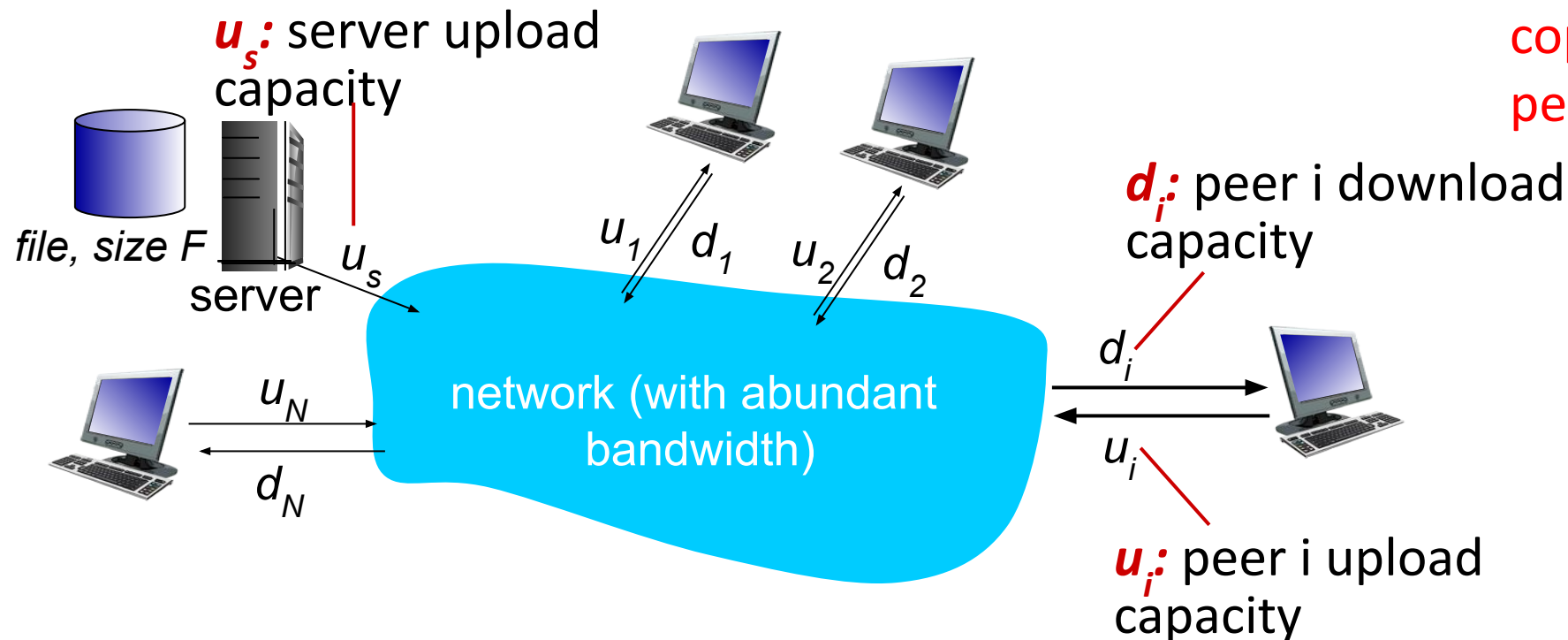
Peer-to-peer (P2P) architecture

- *no* always-on server
- arbitrary end systems directly communicate
- peers request service from other peers, provide service in return to other peers
 - *self scalability* – new peers bring new service capacity, and new service demands
- peers are intermittently connected and change IP addresses
 - complex management
- examples: P2P file sharing (BitTorrent), media streaming (Spotify), VoIP (Skype)



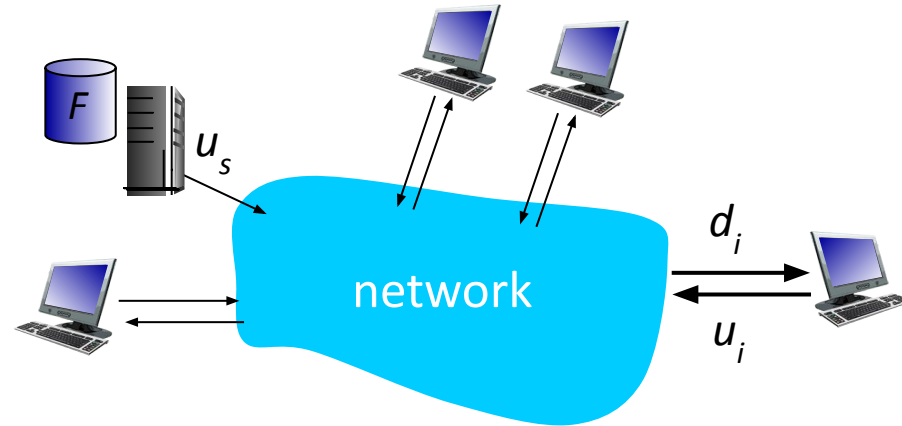
Q: how much time to distribute file (size F) from one server to N peers?

- peer upload/download capacity is limited resource



The **distribution time** is the time it takes to get a copy of the file to all N peers.

- **server transmission:** must sequentially send (upload) N file copies:
 - time to send one copy: F/u_s
 - time to send N copies: NF/u_s
- **client:** each client must download file copy
 - d_{min} = min client download rate =
i.e., $d_{min} = \min \{d_1, d_2, \dots, d_N\}$
 - min client download time: F/d_{min}



time to distribute F
to N clients using
client-server approach $D_{c-s} > \max\{NF/u_s, F/d_{min}\}$

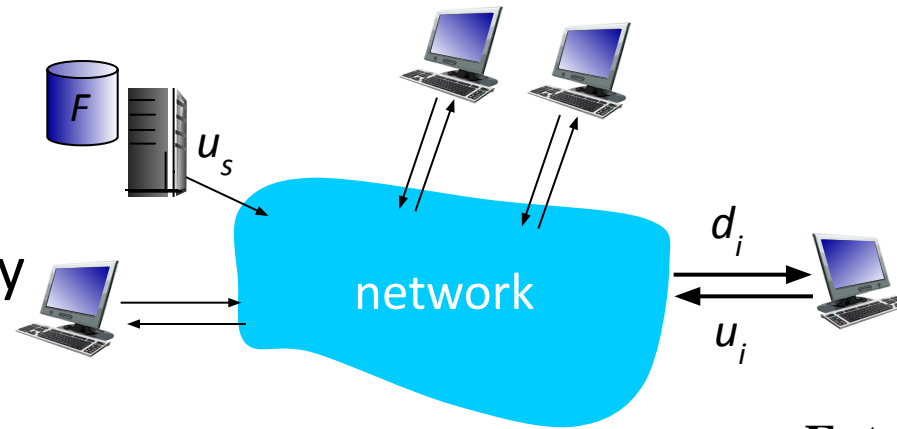
increases linearly in N

- **server transmission:** must upload at least one copy:

- time to send one copy: F/u_s

- **client:** each client must download file copy

- min client download time: F/d_{min}



- **clients:** as aggregate must download NF bits

- max upload rate (limiting max download rate) is

$$u_s + \sum u_i$$

Total upload capacity of the system as a whole

Eqtn - provides a lower bound for the minimum distribution time for the P2P architecture.

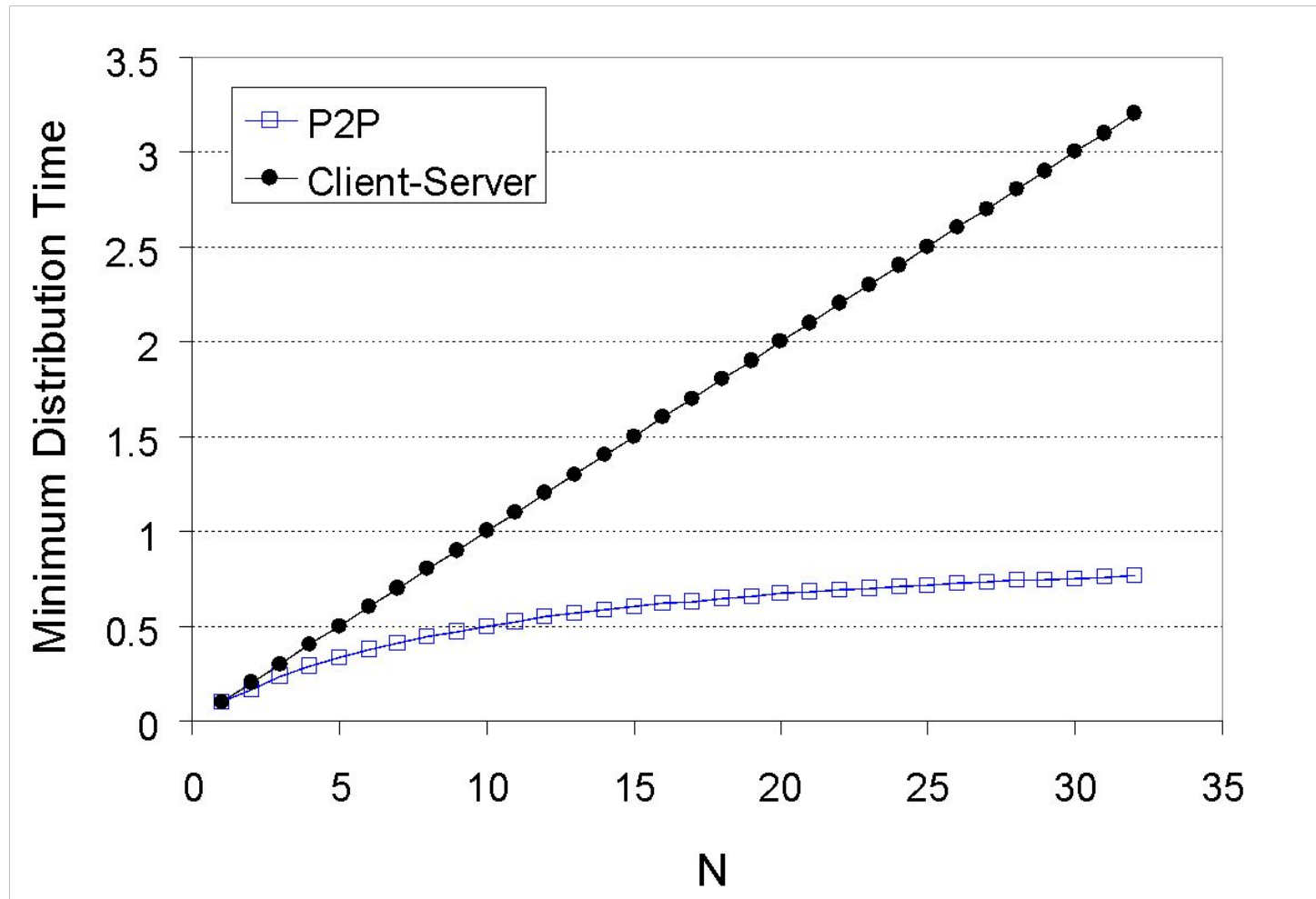
time to distribute F
to N clients using
P2P approach

$$D_{P2P} > \max\{F/u_s, F/d_{min}, NF/(u_s + \sum u_i)\}$$

increases linearly in N

... but so does this, as each peer brings service capacity

Client (all peers) upload rate = u , $F/u = 1$ hour, $u_s = 10u$, $d_{min} \geq u_s$

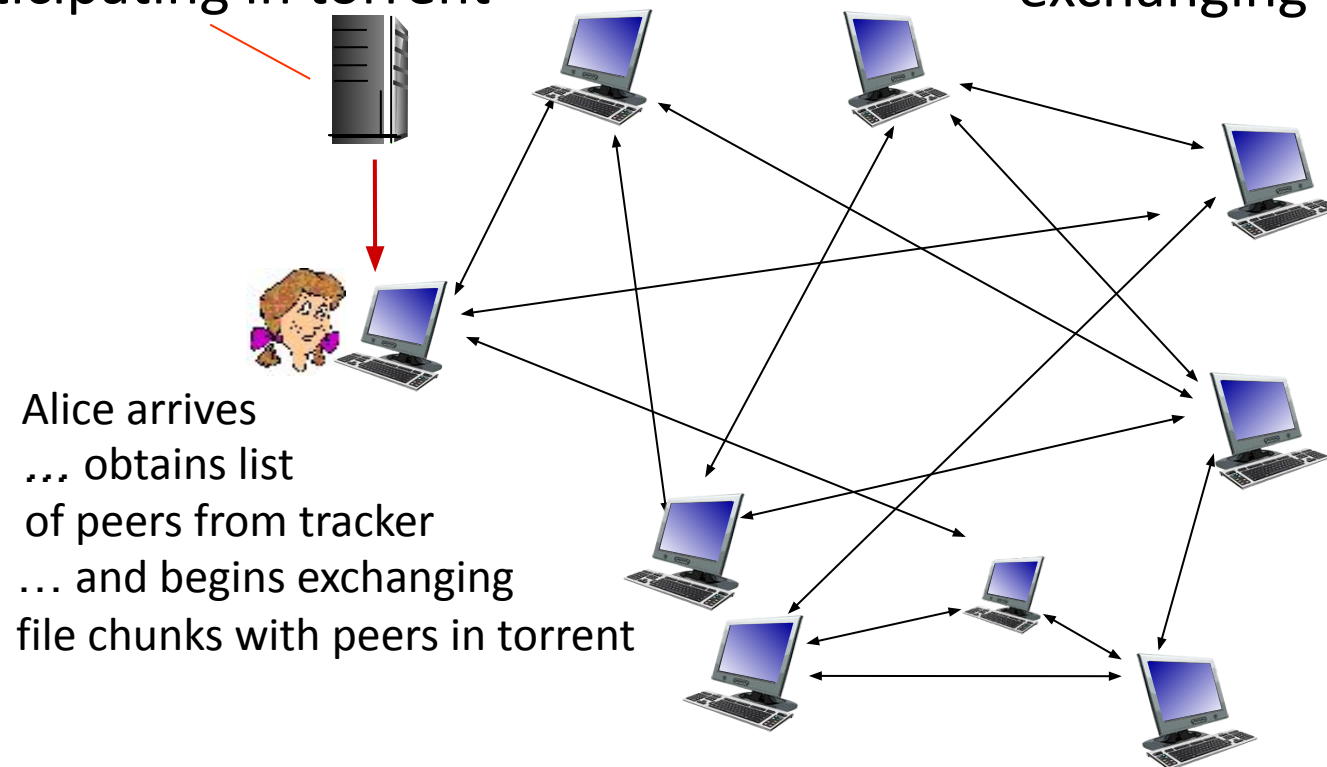


- A peer can transmit the entire file in one hour.
- The server transmission rate is 10 times the peer upload rate.
- Peer download rates are set large enough so as not to have an effect.

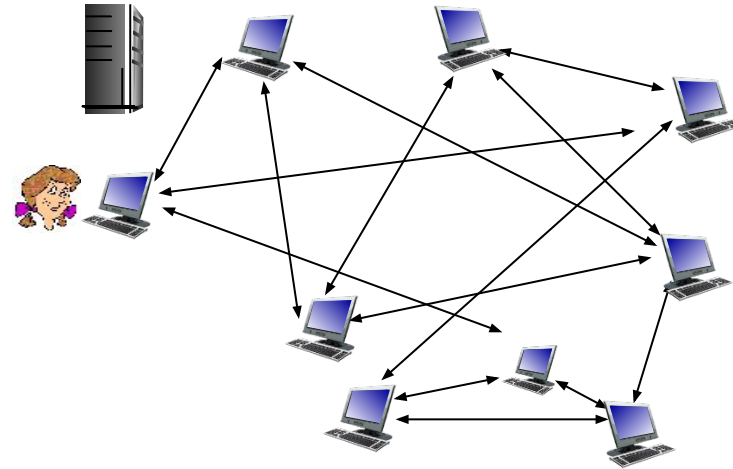
- file divided into 256Kb chunks
- peers in torrent send/receive file chunks

tracker: tracks peers participating in torrent

torrent: group of peers exchanging chunks of a file



- peer joining torrent:
 - has no chunks, but will accumulate them over time from other peers
 - registers with tracker to get list of peers, connects to subset of peers (“neighbors”)
- while downloading, peer uploads chunks to other peers
- peer may change peers with whom it exchanges chunks
- *churn*: peers may come and go
- once peer has entire file, it may (selfishly) leave or (altruistically) remain in torrent



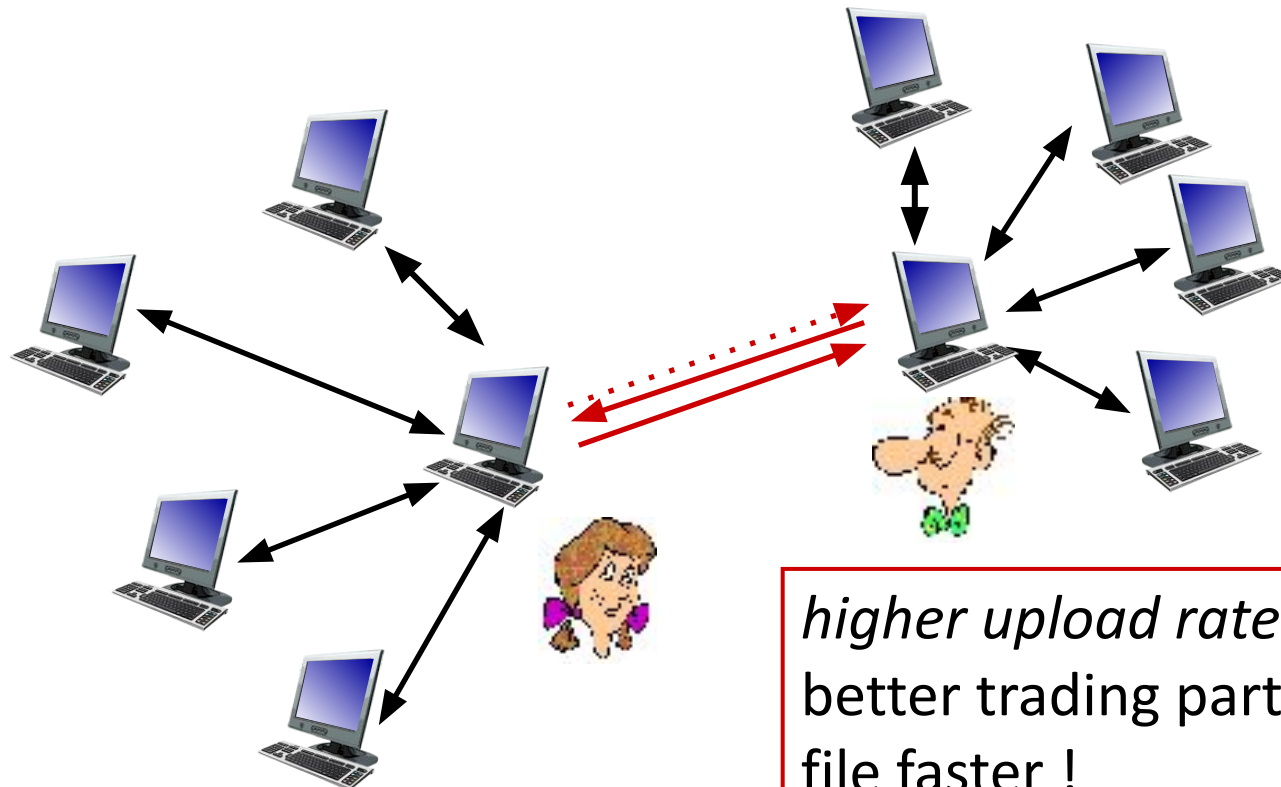
Requesting chunks:

- at any given time, different peers have different subsets of file chunks
- periodically, Alice asks each peer for list of chunks that they have
- Alice requests missing chunks from peers, **rarest first**

Sending chunks: tit-for-tat

- Alice sends chunks to those four peers currently sending her chunks *at highest rate*
 - other peers are choked by Alice (do not receive chunks from her)
 - re-evaluate top 4 every 10 secs
- every 30 secs: randomly select another peer, starts sending chunks
 - “**optimistically unchoke**” this peer
 - newly chosen peer may join top 4

- (1) Alice “optimistically unchokes” Bob
- (2) Alice becomes one of Bob’s top-four providers; Bob reciprocates
- (3) Bob becomes one of Alice’s top-four providers



**Pieces (mini-chunks),
pipelining, random first
selection, endgame mode,
and anti-snubbing**

*higher upload rate: find
better trading partners, get
file faster !*

- BitTorrent (BTT) White Paper –
[https://www.bittorrent.com/btt/btt-docs/BitTorrent \(BTT\) White Paper v0.8.7 Feb 2019.pdf](https://www.bittorrent.com/btt/btt-docs/BitTorrent_(BTT)_White_Paper_v0.8.7_Feb_2019.pdf)
- Peer-to-peer networking with BitTorrent –
<http://web.cs.ucla.edu/classes/cs217/05BitTorrent.pdf>
- Torrents Explained: How BitTorrent Works –
<https://youtu.be/urzQeD7ftbl>



■ Thank You
For Your Attention



THANK YOU

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